

FEA HW2

Ben Sarfati 941180069

April 24 2023

1

Given

$$\begin{cases} -\frac{d}{dx}\left(EA\frac{du}{dx}\right) = \rho A\omega^2 x \\ u(0) = 0 \\ EA\frac{du}{dx}\Big|_{x=L} = 0 \\ -\frac{d}{dx}\left(EA\frac{du}{dx}\right) = \rho A\omega^2 x \end{cases}$$

Integrating and applying boundary conditions twice over:

$$\begin{aligned} \Rightarrow -(EA\frac{du}{dx}) &= \rho A\omega^2 \frac{x^2}{2} + \text{const.} \\ EA\frac{du}{dx}\Big|_{x=L} = 0 &\Rightarrow -(EA\frac{du}{dx}) = \rho A\omega^2 \frac{x^2}{2} - \rho A\omega^2 \frac{L^2}{2} \\ \Rightarrow u &= \frac{\rho\omega^2}{2E} \left((-\frac{1}{3})x^3 + (L^2)x \right) + \text{const.} \\ u(0) = 0 &\Rightarrow u = \frac{\rho\omega^2}{2E} \left((-\frac{1}{3})x^3 + (L^2)x \right) \\ u(x) &= \frac{\rho\omega^2 x}{2E} \left(L^2 - \frac{x^2}{3} \right) \end{aligned}$$

2

Before applying the Galerkin method we define the residuals in the domain and on the Neumann boundary, introducing an approximate form of the solution $\tilde{u} = \phi_j a_j$:

$$\begin{aligned} R &= -\frac{d}{dx}\left(EA\frac{d\tilde{u}}{dx}\right) - \rho A\omega^2 x = -\frac{d}{dx}\left(EA\frac{d\phi_j}{dx}\right)a_j - \rho A\omega^2 x \\ r &= EA\frac{d\tilde{u}}{dx}\Big|_{x=L} = EA\frac{d\phi_j}{dx}a_j\Big|_{x=L} \end{aligned}$$

Now we demand

$$\begin{aligned} \int_0^L \phi_i R dx + \phi_i(L)r &= 0 \\ \Rightarrow \int_0^L \phi_i \left[-\frac{d}{dx}\left(EA\frac{d\phi_j}{dx}\right)a_j - \rho A\omega^2 x \right] dx + \phi_i(L) \left[EA\frac{d\phi_j}{dx}a_j \right]_{x=L} &= 0 \\ \Rightarrow \int_0^L \phi_i \left[-\frac{d}{dx}\left(EA\frac{d\phi_j}{dx}\right)a_j \right] dx + \phi_i(L) \left[EA\frac{d\phi_j}{dx}a_j \right]_{x=L} &= \int_0^L \phi_i \left[\rho A\omega^2 x \right] dx \\ \phi_i \left[-\frac{d}{dx}\left(EA\frac{d\phi_j}{dx}\right)a_j \right] &= \frac{d}{dx} \left[-\phi_i \left(EA\frac{d\phi_j}{dx} \right) a_j \right] + \frac{d\phi_i}{dx} \left(EA\frac{d\phi_j}{dx} \right) a_j \end{aligned}$$

$$\Rightarrow \int_0^L \frac{d\phi_i}{dx} \left(EA \frac{d\phi_j}{dx} \right) a_j dx + \left[-\phi_i \left(EA \frac{d\phi_j}{dx} \right) a_j \right]_0^L + \phi_i(L) \left[EA \frac{d\phi_j}{dx} a_j \right]_{x=L} = \int_0^L \phi_i \left[\rho A \omega^2 x \right] dx$$

We can see that the right boundary terms cancel out which is expected considering a homogeneous Neumann BC. We also demand $i \neq 1 \Rightarrow \phi_i(0) = 0$ in order to get rid of the left boundary term $\forall i > 1$. We are left with

$$i \neq 1 \Rightarrow \int_0^L \frac{d\phi_i}{dx} \left(EA \frac{d\phi_j}{dx} \right) a_j dx = \int_0^L \phi_i \left[\rho A \omega^2 x \right] dx$$

This is the weak form of the problem.

3

We would now like to generalize the problem to a master element. We introduce N nodal points x_j between which lie $n = N - 1$ regions called elements, and the following transformation $\forall e \leq n \in \mathbb{N}$ (e is not in Einstein notation):

$$\xi^e(x) = \frac{2}{h^e} x - \frac{x_e + x_{e+1}}{h^e}; \quad x(\xi^e) = \frac{h^e}{2} \xi^e + \frac{x_e + x_{e+1}}{2}$$

Where

$$h^e = x_{e+1} - x_e$$

$$J^e = \frac{dx}{d\xi^e} = \frac{h^e}{2}$$

For each element there are only two shape functions so we will define

$$\begin{cases} \phi_1^e(x) = \phi_e(x) \\ \phi_2^e(x) = \phi_{e+1}(x) \end{cases}$$

$$\begin{cases} \hat{\phi}_1(\xi^e) = \frac{1}{2}(1 - \xi^e) \\ \hat{\phi}_2(\xi^e) = \frac{1}{2}(1 + \xi^e) \end{cases}$$

$$\begin{cases} B_1(\xi^e) = \frac{d\hat{\phi}_1(\xi^e)}{d\xi^e} = -\frac{1}{2} \\ B_2(\xi^e) = \frac{d\hat{\phi}_2(\xi^e)}{d\xi^e} = \frac{1}{2} \end{cases}$$

Note that

$$\phi_1^e(x(\xi^e)) = \hat{\phi}_1(\xi^e)$$

$$\phi_2^e(x(\xi^e)) = \hat{\phi}_2(\xi^e)$$

Therefore

$$\left. \frac{d\phi_i^e(x)}{dx} \right|_{x(\xi^e)} = \frac{d\phi_i^e(x(\xi^e))}{d\xi^e} \frac{d\xi^e}{dx} = \frac{d\hat{\phi}_i(\xi^e)}{d\xi^e} \frac{1}{J^e} = B_i(\xi^e) \frac{2}{h^e}$$

The integral over the domain can be split into a sum of integrals over each element, and since the only nonzero shape functions over an element are its own, we can say

$$K_{ij}^g = \int_0^L \frac{d\phi_i}{dx} \left(EA \frac{d\phi_j}{dx} \right) dx$$

$$K_{ij}^e = \int_{x_e}^{x_{e+1}} \frac{d\phi_i^e}{dx} \left(EA \frac{d\phi_j^e}{dx} \right) dx$$

$$\mathbf{K}^g = \sum_{e=1}^n \mathbf{A} \mathbf{K}^e$$

Where A is the special matrix summation specific in tutorial 3 of the course. To finish writing the integral form of the stiffness matrix for a master element we simplify to

$$K_{ij}^e = \int_{-1}^1 \frac{1}{J^e} B_i EA \frac{1}{J^e} B_j J^e d\xi^e = \frac{2}{h^e} \int_{-1}^1 B_i EA B_j d\xi^e$$

4

We would now like to calculate the above matrix explicitly using Gaussian quadrature. Since our integrand is a constant function we require $2p - 1 \geq 0 \Rightarrow \mathbb{N} \ni p \geq 1$ Gauss points for exact integral value (we could also simply instruct a solver to multiply the integrand by the length of the master element).

$$\xi_k^e = 0; w_k = 2$$

$$\begin{aligned} K_{ij}^e &= \sum_{k=1}^1 w_k \frac{2}{h^e} B_i E A B_j \\ \Rightarrow \mathbf{K}^e &= \frac{EA}{h^e} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \end{aligned}$$

5

To write the integral form of the forcing vector we introduce a Lagrange polynomial interpolation which exactly represents the linear forcing function:

$$\begin{aligned} \begin{cases} x_1^{\mathcal{L}} = 0 \\ x_2^{\mathcal{L}} = L \end{cases} &; \begin{cases} \mathcal{L}_1 = \frac{x-L}{-L} \\ \mathcal{L}_2 = \frac{x}{L} \end{cases} &; f(x) = \rho A \omega^2 x; f(x) = f(x_j^{\mathcal{L}}) \mathcal{L}_j(x) \\ \int_0^L \phi_i [\rho A \omega^2 x] dx &= \left[\int_0^L \phi_i \mathcal{L}_j(x) dx \right] f(x_j^{\mathcal{L}}) \end{aligned}$$

This process is a bit redundant, and defeats the purpose of the master element transformation. Alternatively, we could introduce the below right-hand side function approximation which does not introduce additional error to the solution:

$$\int_0^L \phi_i [\rho A \omega^2 x] dx \approx \left[\int_0^L \phi_i \phi_j dx \right] \rho A \omega^2 x_j$$

We will work with both expressions since it's unclear which one is the intended way to solve.

6

To calculate the "mass matrix" we apply the same process with the same reasoning as we do to the stiffness matrix from before:

$$\begin{aligned} M_{ij}^g &= \int_0^L \phi_i \mathcal{L}_j(x) dx \\ M_{ij}^e &= \int_{x_e}^{x_{e+1}} \phi_i^e \mathcal{L}_j(x) dx \\ \mathbf{M}^g &= \sum_{e=1}^n \mathbf{A}^e \mathbf{M}^e \\ M_{ij}^e &= \int_{-1}^1 \hat{\phi}_i(\xi^e) \mathcal{L}_j(x(\xi^e)) J^e d\xi^e = \frac{h^e}{2} \int_{-1}^1 \hat{\phi}_i(\xi^e) \mathcal{L}_j(x(\xi^e)) d\xi^e \end{aligned}$$

For Gaussian quadrature we need $2p - 1 \geq 2 \Rightarrow \mathbb{N} \ni p \geq 2$ Gauss points since our integrand is quadratic:

$$\xi_1^e = \frac{1}{\sqrt{3}}; \xi_2^e = -\frac{1}{\sqrt{3}}; w_k = 1$$

$$M_{ij}^e = \sum_{k=1}^2 w_k \hat{\phi}_i(\xi_k^e) \mathcal{L}_j(x(\xi_k^e))$$

$$M_{11}^e = \sum_{k=1}^2 w_k \hat{\phi}_1(\xi_k^e) \mathcal{L}_1(x(\xi_k^e)) = \frac{1}{2} \left(1 - \frac{1}{\sqrt{3}} \right) \left(\frac{\frac{h^e}{2} \frac{1}{\sqrt{3}} + \frac{x_e + x_{e+1}}{2} - L}{-L} \right) + \frac{1}{2} \left(1 + \frac{1}{\sqrt{3}} \right) \left(\frac{-\frac{h^e}{2} \frac{1}{\sqrt{3}} + \frac{x_e + x_{e+1}}{2} - L}{-L} \right)$$

$$\begin{aligned}
&= \frac{\frac{x_e+x_{e+1}}{2} - L}{-L} - \frac{1}{\sqrt{3}} \frac{\frac{h^e}{2} \frac{1}{\sqrt{3}}}{-L} = -\frac{1}{L} \left(\frac{x_e+x_{e+1}}{2} - L - \frac{h^e}{6} \right) \\
M_{12}^e &= \sum_{k=1}^2 w_k \hat{\phi}_1(\xi_k^e) \mathcal{L}_2(x(\xi_k^e)) = \frac{1}{2} \left(1 - \frac{1}{\sqrt{3}} \right) \left(\frac{\frac{h^e}{2} \frac{1}{\sqrt{3}} + \frac{x_e+x_{e+1}}{2}}{L} \right) + \frac{1}{2} \left(1 + \frac{1}{\sqrt{3}} \right) \left(\frac{-\frac{h^e}{2} \frac{1}{\sqrt{3}} + \frac{x_e+x_{e+1}}{2}}{L} \right) \\
&= \frac{\frac{x_e+x_{e+1}}{2}}{L} - \frac{1}{\sqrt{3}} \frac{\frac{h^e}{2} \frac{1}{\sqrt{3}}}{L} = \frac{1}{L} \left(\frac{x_e+x_{e+1}}{2} - \frac{h^e}{6} \right) \\
M_{21}^e &= \sum_{k=1}^2 w_k \hat{\phi}_2(\xi_k^e) \mathcal{L}_1(x(\xi_k^e)) = \frac{1}{2} \left(1 + \frac{1}{\sqrt{3}} \right) \left(\frac{\frac{h^e}{2} \frac{1}{\sqrt{3}} + \frac{x_e+x_{e+1}}{2} - L}{-L} \right) + \frac{1}{2} \left(1 - \frac{1}{\sqrt{3}} \right) \left(\frac{-\frac{h^e}{2} \frac{1}{\sqrt{3}} + \frac{x_e+x_{e+1}}{2} - L}{-L} \right) \\
&= \frac{\frac{x_e+x_{e+1}}{2} - L}{-L} + \frac{1}{\sqrt{3}} \frac{\frac{h^e}{2} \frac{1}{\sqrt{3}}}{-L} = -\frac{1}{L} \left(\frac{x_e+x_{e+1}}{2} - L + \frac{h^e}{6} \right) \\
M_{22}^e &= \sum_{k=1}^2 w_k \hat{\phi}_2(\xi_k^e) \mathcal{L}_2(x(\xi_k^e)) = \frac{1}{2} \left(1 + \frac{1}{\sqrt{3}} \right) \left(\frac{\frac{h^e}{2} \frac{1}{\sqrt{3}} + \frac{x_e+x_{e+1}}{2}}{L} \right) + \frac{1}{2} \left(1 - \frac{1}{\sqrt{3}} \right) \left(\frac{-\frac{h^e}{2} \frac{1}{\sqrt{3}} + \frac{x_e+x_{e+1}}{2}}{L} \right) \\
&= \frac{\frac{x_e+x_{e+1}}{2}}{L} + \frac{1}{\sqrt{3}} \frac{\frac{h^e}{2} \frac{1}{\sqrt{3}}}{L} = \frac{1}{L} \left(\frac{x_e+x_{e+1}}{2} + \frac{h^e}{6} \right) \\
\Rightarrow \mathbf{M}^e &= \frac{1}{L} \begin{bmatrix} -\left(\frac{x_e+x_{e+1}}{2} - L - \frac{h^e}{6} \right) & \left(\frac{x_e+x_{e+1}}{2} - \frac{h^e}{6} \right) \\ -\left(\frac{x_e+x_{e+1}}{2} - L + \frac{h^e}{6} \right) & \left(\frac{x_e+x_{e+1}}{2} + \frac{h^e}{6} \right) \end{bmatrix}
\end{aligned}$$

The global forcing function is:

$$\mathbf{f} = \mathbf{f}^e = \begin{bmatrix} 0 \\ \rho A \omega^2 L \end{bmatrix}$$

It is the same as the local forcing function in this case because the Lagrange polynomials are global. The local forcing vector is then

$$\mathbf{F}^e = \mathbf{M}^e \mathbf{f}^e = \rho A \omega^2 \begin{bmatrix} \left(\frac{x_e+x_{e+1}}{2} - \frac{h^e}{6} \right) \\ \left(\frac{x_e+x_{e+1}}{2} + \frac{h^e}{6} \right) \end{bmatrix}$$

Alternatively, with the alternative RHS approximation:

$$\begin{aligned}
M_{ij}^g &= \int_0^L \phi_i \phi_j dx \\
M_{ij}^e &= \int_{x_e}^{x_{e+1}} \phi_i^e \phi_j^e dx \\
\mathbf{M}^g &= \sum_{e=1}^n \mathbf{M}^e \\
M_{ij}^e &= \int_{-1}^1 \hat{\phi}_i(\xi^e) \hat{\phi}_j(\xi^e) J^e d\xi^e = \frac{h^e}{2} \int_{-1}^1 \hat{\phi}_i(\xi^e) \hat{\phi}_j(\xi^e) d\xi^e
\end{aligned}$$

Once again for Gaussian quadrature we need $2p-1 \geq 2 \Rightarrow \mathbb{N} \ni p \geq 2$ Gauss points since our integrand is quadratic:

$$\begin{aligned}
\xi_1^e &= \frac{1}{\sqrt{3}}; \quad \xi_2^e = -\frac{1}{\sqrt{3}}; \quad w_k = 1 \\
\frac{2}{h^e} M_{ij}^e &= \sum_{k=1}^2 w_k \hat{\phi}_i(\xi_k^e) \hat{\phi}_j(\xi_k^e) \\
\frac{2}{h^e} M_{11}^e &= \sum_{k=1}^2 w_k \hat{\phi}_1(\xi_k^e) \hat{\phi}_1(\xi_k^e) = \frac{1}{4} \left(1 - \frac{1}{\sqrt{3}} \right)^2 + \frac{1}{4} \left(1 + \frac{1}{\sqrt{3}} \right)^2 = \frac{1}{4} \left(1 - 2 \frac{1}{\sqrt{3}} + \frac{1}{3} \right) + \frac{1}{4} \left(1 + 2 \frac{1}{\sqrt{3}} + \frac{1}{3} \right) = \frac{2}{3} \\
\frac{2}{h^e} M_{12}^e &= \sum_{k=1}^2 w_k \hat{\phi}_1(\xi_k^e) \hat{\phi}_2(\xi_k^e) = \frac{1}{4} \left(1 - \frac{1}{\sqrt{3}} \right) \left(1 + \frac{1}{\sqrt{3}} \right) + \frac{1}{4} \left(1 + \frac{1}{\sqrt{3}} \right) \left(1 - \frac{1}{\sqrt{3}} \right) = \frac{1}{3}
\end{aligned}$$

$$\begin{aligned}\frac{2}{h^e} M_{21}^e &= \sum_{k=1}^2 w_k \hat{\phi}_1(\xi_k^e) \hat{\phi}_2(\xi_k^e) = \frac{1}{4} \left(1 + \frac{1}{\sqrt{3}}\right) \left(1 - \frac{1}{\sqrt{3}}\right) + \frac{1}{4} \left(1 - \frac{1}{\sqrt{3}}\right) \left(1 + \frac{1}{\sqrt{3}}\right) = \frac{1}{3} \\ \frac{2}{h^e} M_{22}^e &= \sum_{k=1}^2 w_k \hat{\phi}_1(\xi_k^e) \hat{\phi}_1(\xi_k^e) = \frac{1}{4} \left(1 + \frac{1}{\sqrt{3}}\right)^2 + \frac{1}{4} \left(1 - \frac{1}{\sqrt{3}}\right)^2 = \frac{1}{4} \left(1 + 2\frac{1}{\sqrt{3}} + \frac{1}{3}\right) + \frac{1}{4} \left(1 - 2\frac{1}{\sqrt{3}} + \frac{1}{3}\right) = \frac{2}{3} \\ &\Rightarrow \mathbf{M}^e = \frac{h^e}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\end{aligned}$$

The global forcing function is simply $f_j = \rho A \omega^2 x_j$ so the local forcing function is

$$\mathbf{f}^e = \rho A \omega^2 \begin{bmatrix} x_e \\ x_{e+1} \end{bmatrix}$$

So the local forcing vector is

$$\mathbf{F}^e = \mathbf{M}^e \mathbf{f}^e = \rho A \omega^2 \frac{h^e}{6} \begin{bmatrix} 2x_e + x_{e+1} \\ x_e + 2x_{e+1} \end{bmatrix}$$

7

Solving with $n = 2$:

$$\Rightarrow h^e = \frac{L}{n} = \frac{L}{2}; \quad x_j = (j-1)h^e = (j-1)\frac{L}{2}$$

$$\mathbf{K}^g \mathbf{a} = \mathbf{F}^g$$

$$\mathbf{K}^e = \frac{EA}{h^e} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\Rightarrow \mathbf{K}^g = \frac{EA}{h^e} \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\mathbf{F}^e = \rho A \omega^2 \frac{h^e}{6} \begin{bmatrix} 2x_e + x_{e+1} \\ x_e + 2x_{e+1} \end{bmatrix}$$

$$\Rightarrow \mathbf{F}^g = \rho A \omega^2 \frac{h^e}{6} \begin{bmatrix} 2x_1 + x_2 \\ x_1 + 4x_2 + x_3 \\ x_2 + 2x_3 \end{bmatrix}$$

Note that the system is incorrect as long as the dirichlet BC is not present:

$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \frac{\rho(h^e \omega)^2}{6E} \begin{bmatrix} 0 \\ x_1 + 4x_2 + x_3 \\ x_2 + 2x_3 \end{bmatrix}$$

Moving the first column to the RHS:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \frac{\rho(h^e \omega)^2}{6E} \begin{bmatrix} 0 \\ x_1 + 4x_2 + x_3 \\ x_2 + 2x_3 \end{bmatrix} - 0 \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$$

Simplifying:

$$\begin{aligned} \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} a_2 \\ a_3 \end{bmatrix} &= \frac{\rho(h^e \omega)^2}{6E} \begin{bmatrix} x_1 + 4x_2 + x_3 \\ x_2 + 2x_3 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} a_2 \\ a_3 \end{bmatrix} &= \frac{\rho(h^e \omega)^2}{6E} \begin{bmatrix} x_1 + 5x_2 + 3x_3 \\ x_2 + 2x_3 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a_2 \\ a_3 \end{bmatrix} &= \begin{bmatrix} a_2 \\ a_3 \end{bmatrix} = \frac{\rho(h^e \omega)^2}{6E} \begin{bmatrix} x_1 + 5x_2 + 3x_3 \\ x_1 + 6x_2 + 5x_3 \end{bmatrix} = \frac{\rho(h^e \omega)^2 L}{6E} \begin{bmatrix} \frac{11}{2} \\ 8 \end{bmatrix} \end{aligned}$$

Entering values:

$$\begin{cases} a_1 = 0 \\ a_2 = 2.8408 \cdot 10^{-6} \\ a_3 = 4.1321 \cdot 10^{-6} \end{cases}$$

Finally, our choice of shape functions allows us to make the simplification

$$\tilde{u} = \phi_j a_j \Rightarrow \tilde{u} = \begin{cases} \phi_j^e a_{e+j-1} & x_e \leq x \leq x_{e+1} \end{cases}$$

Again, this applies $\forall e$; e is not in Einstein notation.

$$\phi_1^1(x) = \hat{\phi}_1(\xi^1(x)) = \frac{1}{2} \left(1 - \left(\frac{2}{h^e} x - \frac{x_1 + x_2}{h^e} \right) \right) = 1 - x$$

$$\phi_2^1(x) = \hat{\phi}_2(\xi^1(x)) = \frac{1}{2} \left(1 + \left(\frac{2}{h^e} x - \frac{x_1 + x_2}{h^e} \right) \right) = x$$

$$\phi_1^2(x) = \hat{\phi}_1(\xi^2(x)) = \frac{1}{2} \left(1 - \left(\frac{2}{h^e} x - \frac{x_2 + x_3}{h^e} \right) \right) = 2 - x$$

$$\phi_2^2(x) = \hat{\phi}_2(\xi^2(x)) = \frac{1}{2} \left(1 + \left(\frac{2}{h^e} x - \frac{x_2 + x_3}{h^e} \right) \right) = x - 1$$

$$\Rightarrow \tilde{u} = \begin{cases} (2.8408 \cdot 10^{-6})x & 0 \leq x \leq \frac{L}{2} \\ (2.8408 \cdot 10^{-6})(2 - x) + (4.1321 \cdot 10^{-6})(x - 1) & \frac{L}{2} \leq x \leq L \end{cases}$$

8

See MATLAB code.

9

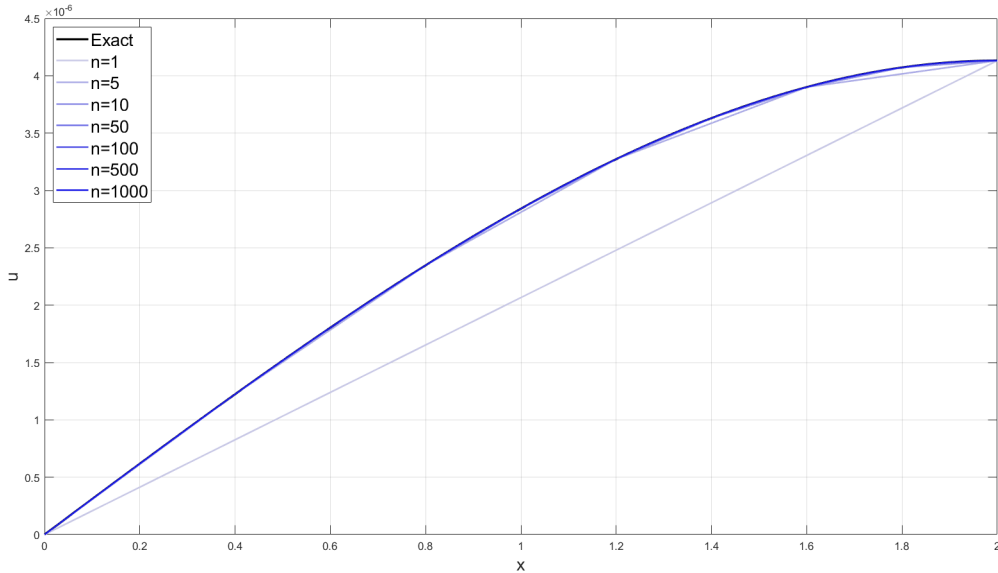


Figure 1: Q9: Exact Solution vs. Numerical Solution Using 1, 5, 10, 100 Elements

The exact solution is difficult to see because it's actually hidden behind the 100 element numerical solution, which is a good sign considering that a rapid convergence to the exact solution as number of elements increases is expected.

10

To aid the calculations of L_2 and H_1 the Gaussian quadrature was done "on paper:"

$$\begin{aligned} L_2 &= \sqrt{\int_{\Omega} (u - \tilde{u})^2 dx} = \sqrt{\sum_{e=1}^n \int_{x_e}^{x_{e+1}} (u(x) - \tilde{u}(x))^2 dx} \\ &= \sqrt{\sum_{e=1}^n \int_{-1}^1 \left(u(x(\xi^e)) - \tilde{u}(x(\xi^e)) \right)^2 |J^e| d\xi^e} \end{aligned}$$

Using $\tilde{u} = \phi_j a_j \Rightarrow \tilde{u} = \left\{ \phi_j^e a_{e+j-1} \quad x_e \leq x \leq x_{e+1} \right\}$:

$$\tilde{u}(x(\xi^e)) = a_{e+j-1} \phi_j^e(x(\xi^e)) = a_{e+j-1} \hat{\phi}_j(\xi^e)$$

We can make the simplification because we are only evaluating this term within the bounds of the integral, which are the bounds of the element.

If we only want to use 1 Gauss point:

$$\xi_k^e = 0; \quad w_k = 2$$

$$\begin{aligned} L_2 &\approx \sqrt{\sum_{e=1}^n 2 \left(u(x(0)) - a_{e+j-1} \hat{\phi}_j(0) \right)^2 |J^e|} \\ L_2 &\approx \sqrt{h^e \sum_{e=1}^n \left(u\left(\frac{x_e + x_{e+1}}{2}\right) - \frac{a_e + a_{e+1}}{2} \right)^2} \end{aligned}$$

As for H_1 :

$$\begin{aligned} H_1 &= \sqrt{\int_{\Omega} \left((u - \tilde{u})^2 + \left(\frac{du}{dx} - \frac{d\tilde{u}}{dx} \right)^2 \right) dx} = \sqrt{\sum_{e=1}^n \int_{x_e}^{x_{e+1}} \left((u(x) - \tilde{u}(x))^2 + \left(\frac{du(x)}{dx} - \frac{d\tilde{u}(x)}{dx} \right)^2 \right) dx} \\ &= \sqrt{\sum_{e=1}^n \int_{-1}^1 \left[\left(u(x(\xi^e)) - \tilde{u}(x(\xi^e)) \right)^2 + \left(\frac{du(x)}{dx} \Big|_{x(\xi^e)} - \frac{d\tilde{u}(x)}{dx} \Big|_{x(\xi^e)} \right)^2 \right] |J^e| d\xi^e} \end{aligned}$$

Again, since we are only evaluating these two $\tilde{u}$ -terms within the bounds of the integral, we can say:

$$\tilde{u}(x) = a_{e+j-1} \phi_j^e(x)$$

$$\frac{d\tilde{u}(x)}{dx} = a_{e+j-1} \frac{d\phi_j^e(x)}{dx}$$

And if we want to evaluate these expressions in the local domain:

$$\begin{aligned} a_{e+j-1} \phi_j^e(x(\xi^e)) &= a_{e+j-1} \hat{\phi}_j(\xi^e) \\ a_{e+j-1} \frac{d\phi_j^e(x)}{dx} \Big|_{x(\xi^e)} &= a_{e+j-1} \frac{d\phi_j^e(x(\xi^e))}{d\xi^e} \frac{d\xi^e}{dx} = a_{e+j-1} \frac{d\hat{\phi}_j(\xi^e)}{d\xi^e} \frac{1}{J^e} = a_{e+j-1} B_j(\xi^e) \frac{2}{h^e} \end{aligned}$$

So

$$H_1 = \sqrt{\sum_{e=1}^n \int_{-1}^1 \left[\left(u(x(\xi^e)) - a_{e+j-1} \hat{\phi}_j(\xi^e) \right)^2 + \left(\frac{du(x)}{dx} \Big|_{x(\xi^e)} - a_{e+j-1} B_j(\xi^e) \frac{2}{h^e} \right)^2 \right] \frac{h^e}{2} d\xi^e}$$

Using 2 Gauss points:

$$\xi_1^e = \frac{1}{\sqrt{3}}; \quad \xi_2^e = -\frac{1}{\sqrt{3}}; \quad w_k = 1$$

$$H_1 = \sqrt{\frac{h^e}{2} \sum_{e=1}^n (I_1 + I_2)}$$

$$I_1 = \left(u\left(x\left(\frac{1}{\sqrt{3}}\right)\right) - a_{e+j-1} \hat{\phi}_j\left(\frac{1}{\sqrt{3}}\right) \right)^2 + \left(u\left(x\left(-\frac{1}{\sqrt{3}}\right)\right) - a_{e+j-1} \hat{\phi}_j\left(-\frac{1}{\sqrt{3}}\right) \right)^2$$

$$I_2 = \left(\left. \frac{du(x)}{dx} \right|_{x(\frac{1}{\sqrt{3}})} - a_{e+j-1} B_j\left(\frac{1}{\sqrt{3}}\right) \frac{2}{h^e} \right)^2 + \left(\left. \frac{du(x)}{dx} \right|_{x(-\frac{1}{\sqrt{3}})} - a_{e+j-1} B_j\left(-\frac{1}{\sqrt{3}}\right) \frac{2}{h^e} \right)^2$$

$$= \left(\left. \frac{du(x)}{dx} \right|_{x(\frac{1}{\sqrt{3}})} - \frac{a_{e+1} - a_e}{h^e} \right)^2 + \left(\left. \frac{du(x)}{dx} \right|_{x(-\frac{1}{\sqrt{3}})} - \frac{a_{e+1} - a_e}{h^e} \right)^2$$

$$u(x) = \frac{\rho\omega^2 x}{2E} \left(L^2 - \frac{x^2}{3} \right) \Rightarrow \frac{du(x)}{dx} = \frac{\rho\omega^2}{2E} (L^2 - x^2)$$

The resulting rates of convergence for each error norm are listed below.

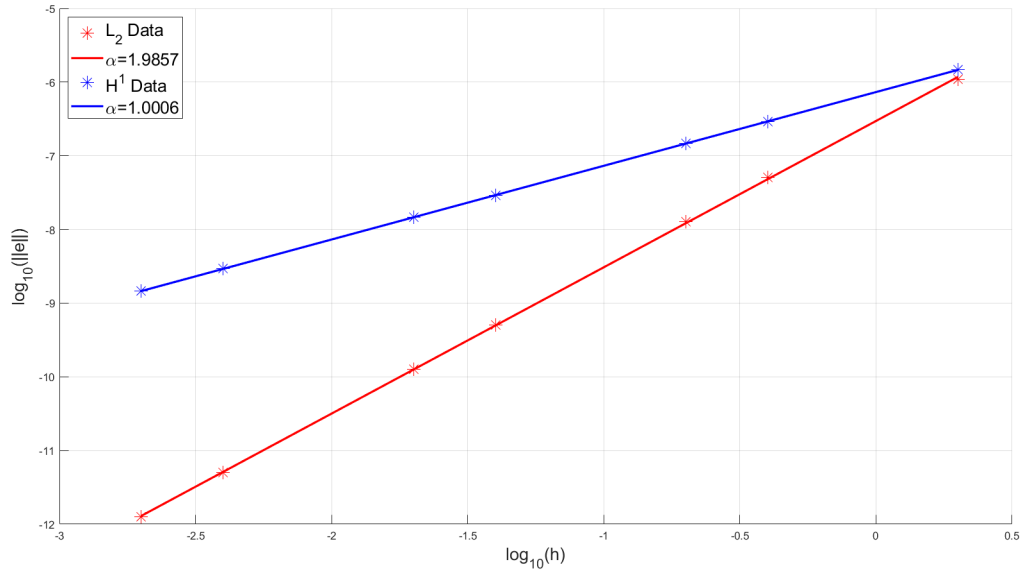


Figure 2: Q10: Logarithmic Error Norms vs. Logarithmic Element Size h

$$L_2 \Rightarrow \alpha = 1.9857 \approx 2$$

$$H_1 \Rightarrow \alpha = 1.0006 \approx 1$$